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Data Analysis Essentials

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Lab Practice

Practice:

Week 2: The Shape and Reshaping of NumPy Array

a. Dimensions of NumPy array
b. Shape of NumPy array
c. Size of NumPy array
d. Reshaping a NumPy array
e. Flattening a NumPy array
f. Transpose of a NumPy array

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ndarray

NumPy provides various attributes and functions to determine the structure of an array and modify its shape without changing the data.

The important concepts are:

Dimensions (ndim)

Shape (shape)

Size (size)

Reshaping (reshape())

Flattening (flatten())

Transpose (transpose() or .T)

array

(a) Dimensions of NumPy Array

The dimension of a NumPy array refers to the number of axes (directions) required to represent the array.

1D Array→ One axis

2D Array→ Two axes (rows and columns)

3D Array→ Three axes

NumPy uses the ndim attribute to find the dimension of an array.

Syntax

array.ndim

One-Dimensional Array

Program

```
import numpy as np
arr = np.array([10, 20, 30, 40])
print("Array:")
print(arr)
print("Dimensions:", arr.ndim)
```

Output

```
Array: [10 20 30 40]
Dimensions: 1
```

Two-Dimensional Array

Program

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6]])
print(arr)
print("Dimensions:", arr.ndim)
```

Output

```
[[1 2 3] [4 5 6]]
Dimensions: 2
```

Three-Dimensional Array

Program

```
import numpy as np
arr = np.array([[[1,2],[3,4]],[[5,6],[7,8]]])
print(arr)
print("Dimensions:", arr.ndim)
```

Output

```
[[[1 2]
[3 4]]
[[5 6]
[7 8]]]
Dimensions: 3
```

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(b) Shape of NumPy Array

The shape of an array tells us how many rows and columns (or elements in each dimension) the array contains.

The shape attribute returns a tuple.

Syntax

array.shape

Program

```
import numpy as np
arr = np.array([[10,20,30], [40,50,60]])
print(arr)
print("Shape:", arr.shape)
```

Output

```
[[10 20 30]
[40 50 60]]
Shape: (2, 3)
```

ndarray

To find 3-D array Shape

Program

```
import numpy as np
arr = np.array([[[1,2],[3,4]], [[5,6],[7,8]]])
print(arr.shape)
```

Output

```
(2, 2, 2)
```

ndarray

(c) Size of NumPy Array

The size of an array is the total number of elements present in it.

NumPy uses the size attribute.

Syntax

array.size

Program

```
import numpy as np
arr = np.array([[10,20,30],[40,50,60]])
print(arr)
print("Size:", arr.size)
```

Output

```
[[10 20 30]
[40 50 60]]
Size: 6
```

ndarray

Program

```
import numpy as np
arr = np.array([1,2,3,4,5,6,7])
print(arr.size)
```

Output

7

2Darray

(d) Reshaping a NumPy Array

Reshaping changes the dimensions of an array without changing its data.The total number of elements must remain the same.

NumPy provides the reshape() function.

Syntax

array.reshape(rows, columns)

Program

```
import numpy as np
arr = np.arange(12)
print("Original Array:")
print(arr)
new_arr = arr.reshape(3,4)
print("\nReshaped Array:")
print(new_arr)
```

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ndarray

Output

Original Array: [0 1 2 3 4 5 6 7 8 9 10 11]

Reshaped Array: [[0 1 2 3]

[4 5 6 7]

[8 9 10 11]]

ndarray

Program

```
import numpy as np
arr = np.arange(16)
new_arr = arr.reshape(4,4)
print(new_arr)
```

Output

```
[[ 0 1 2 3]
[ 4 5 6 7]
[ 8 9 10 11]
[12 13 14 15]]
```

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ndarray

Program

```
import numpy as np
arr = np.arange(12)
new_arr = arr.reshape(3,-1)
print(new_arr)
```

Output

```
[[ 0  1  2  3]
[ 4  5  6  7]
[ 8  9 10 11]]
```

-1 tells NumPy to calculate the number of columns automatically.

ndarray

(e) Flattening a NumPy Array

Flattening converts a multi-dimensional array into a one-dimensional array.NumPy provides two methods: flatten(), ravel()

Syntax

array.flatten()

Program

```
import numpy as np
arr = np.array([[10,20,30],[40,50,60]])
print("Original Array:")
print(arr)
flat = arr.flatten()
print("\nFlattened Array:")
print(flat)
```

Output

```
Original Array: [[10 20 30]
                 [40 50 60]]
Flattened Array: [10 20 30 40
50 60]
```

ndarray

Program

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6]])
print(arr.ravel())
```

Output

```
[1 2 3 4 5 6]
```

Flatten()	Ravel()
Returns a copy	Returns a view whenever possible
Uses more memory	Uses less memory
Original array is not affected	Changes may affect original array

ndarray

(f) Transpose of a NumPy Array

The transpose of a matrix interchanges its rows and columns.NumPy provides: .T, transpose()

Syntax

array.T

np.transpose(array)

Program

```
import numpy as np
arr = np.array([[1,2,3],[4,5,6]])
print("Original Array:")
print(arr)
print("\nTranspose:")
print(arr.T)
```

Output

```
Original Array: [[1 2 3]
                 [4 5 6]]
Transpose: [[1 4]
            [2 5]
            [3 6]]
```

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ndarray

Program

```
import numpy as np
arr = np.array([[10,20], [30,40],[50,60]])
transpose = np.transpose(arr)
print(transpose)
```

Output

```
[[10 30 50]
[20 40 60]]
```

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Thank You